

LegalForecastBench: Forecasting Judicial Decisions as a Test of Legal Reasoning Ability

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Abstract

Reinforcement learning (RL) techniques, in domains that can offer objectively verifiable rewards, have led to incredible and noteworthy recent gains in the capabilities of frontier AI models. But not all domains are equally verifiable. Many high-level legal judgment tasks are difficult to verify, and model capability gains on those tasks have not been nearly as noteworthy as model gains in domains that are susceptible to automated verification like math and coding.

Existing benchmarks used to evaluate high-level legal tasks rely largely on human-written rubrics that test model output in response to fictitious, AI-generated environments. But an AI-assisted audit of one of the most widely used such benchmarks, checked by a practicing litigator's review of a random sample of the flagged criteria, found evidence of hundreds of erroneous rubrics in just one subpart of the benchmark, many of them in task environments that are themselves misspecified. This adds to a growing number of instances in which practicing attorneys have pointed out errors in such benchmarks. The errors in these environments illustrate the challenges facing even the top data providers and research labs trying to improve frontier model capabilities, at a time when many previously challenging benchmarks and tasks have become saturated.

This paper argues that the heavy reliance on artificially constructed environments fails to take advantage of key properties of the common law judicial tradition, which gives precedential effect and the force of law to reasoned judicial decisions issued by judges in specific cases or controversies. For AI researchers, those judicial dispositions supply an external signal that allows for evaluation of some high-level legal judgment tasks against an objectively verifiable real-world ground truth. In the legal system, the opinions of purported "legal experts" or data providers as to what the law is have no authoritative or precedential value. The pronouncements of a tribunal, by contrast, carry the force of law (and often precedential effect).

LegalForecastBench illustrates one mechanism by which the signal from judicial decisions can be used to objectively evaluate the legal reasoning capabilities of frontier models. Models receive the same briefs that a federal judge received when ruling on a motion to dismiss, and are asked to estimate the probability that the judge dismissed each claim challenged by a defendant.

The beta includes 91 cases and 387 scored claim-defendant units. In the initial run, GPT-6 Sol has the lowest micro-Brier loss (0.118) and the highest accuracy (84.5%) of seventeen models tested. Notably, the test showed that almost all of the frontier reasoning models now outperform the retrospectively fitted constant-probability reference (i.e., the base rate), which human lawyers have not been able to outperform in past studies of prediction. We describe how more training signal could potentially be extracted from real-world litigation data, and suggest this approach may be more effective than relying on fictitious environments created by some data providers.

1 Introduction

Reinforcement learning (RL) techniques have led to significant recent gains in frontier model capabilities, but the most noteworthy gains have been in domains like mathematics and coding that are susceptible to automated verification. Reinforcement learning with verifiable rewards (RLVR) has allowed frontier models to gain capabilities by receiving reward signals from automated correctness checks (like mathematical answer matching, executable tests, or formal proof checking) [1, 2]. Earlier self-play systems demonstrated the powerful capabilities that automated outcome feedback could provide to models playing games with explicit rules and automatable scoring mechanisms [3].

Frontier models have also made noteworthy gains on legal tasks that are objectively verifiable; for example, tests that ask models bar exam-style questions about established legal propositions have become largely saturated [13]. This increase in legal capabilities has forced researchers to develop new tasks that challenge today’s frontier models. The literature illustrates at least two approaches being used recently: preference evaluation (comparing which of two outputs practicing attorneys prefer) and rubric-based reinforcement learning (grading model outputs against specified criteria to produce a reward) [7]. While these approaches are important and useful, our audit of one of the most widely used rubric-based legal benchmarks (which utilized rubrics supplied by leading data companies) found that more than 13.4% of the criteria in the litigation practice area (95% CI) contained material errors that could impact the grading on those rubrics, and more than 11.1% come from task environments with errors or inconsistencies. This illustrates the extraordinary difficulty of obtaining reliable, high-quality training data that can improve frontier models’ legal abilities at current capability levels.

This paper aims to demonstrate that judicial dispositions in litigation matters provide a valuable external signal that can make high-level legal judgment tasks verifiable against an objective ground truth. This approach rests on a simple reality of the legal system: "It is emphatically the province and duty of the Judicial Department to say what the law is." *Marbury v. Madison*, 5 U.S. 137, 177 (1803). Every litigated judgment has the force of law as between the parties, and definitively resolves any disputes between the parties on the interpretation of the law or its application to a specific fact pattern.

LegalForecastBench illustrates one specific use of this signal, using judges’ rulings on motions to dismiss to evaluate the predictive abilities of frontier models. The benchmark focuses on district court level motions because lower courts are heavily constrained by precedent and largely decide garden-variety disputes in which the ability to correctly analyze and apply the law to the alleged facts will be the primary way to predict the outcome. Prediction at this level is likely to serve as a reasonable proxy for legal analytical ability (whereas the U.S. Supreme Court is not nearly as constrained by precedent, since it is free to overrule any of its own prior precedential decisions, and prediction at that level would focus on a much greater extent on modeling the preferences or likely vote of individual justices). Obviously, not every judicial disposition is accurate, and there are some politically salient cases even in the lower courts where different judges might have different leanings, so the prediction task is not a pure test of legal reasoning ability.

But even so, frontier models’ prediction ability is valuable and important, because prediction is a high-level legal judgment task often performed by the most senior and highly compensated lawyer on the case team because of its enormous consequences for the client’s legal strategy. In many kinds of cases (such as securities fraud class actions), whether the motion to dismiss is granted or denied can have an enormous impact on the practical economic value of a case; the vast majority of cases in which pre-trial dismissal motions are denied end in settlement, whereas a case that is dismissed at

the outset usually results in no recovery to the plaintiff [21, p. 16]. Prior research shows that even parties who have no intention of litigating negotiate legal resolutions "in the shadow of the law" [22]. When deciding whether to settle a dispute or litigate, each side's expected value from litigation (net of the costs and risks involved) can function as a kind of reservation value; if the settlement offer is worth less than the expected value from litigation, then a party will rationally reject settlement and litigate instead. [52, 53]. This can make settlement difficult or impossible if both sides are overly optimistic about their chances of success in litigation; models that help lawyers and their clients better calibrate their expectations could help unlock the enormous surplus that litigants achieve when they settle consensually (avoiding the costs, risks, and distraction brought on by protracted, adversarial litigation), by helping both sides to see their case more realistically, which should make them more willing to offer and accept the fair value of the case.

2 Related work

2.1 Legal outcome prediction

Using AI models to predict litigation outcomes has a long lineage, although the incredible recent growth in frontier model capabilities has created opportunities for more powerful approaches than those available to prior researchers. One early experiment involved the 2002 Term of the U.S. Supreme Court; Ruger and colleagues predicted the outcome of cases that Term using hand-coded metadata and statistical techniques [35]. Later researchers used more sophisticated machine-learning and natural language processing techniques. Aletras and colleagues in 2016 used a text-based approach to predict the outcomes of European Court of Human Rights decisions [37]. Katz, Bommarito, and Blackman in 2017 used a random-forest classifier over structured Supreme Court data to predict Justice votes and case outcomes. [36]. CAIL2018 used a dataset of more than 2.6 million Chinese criminal cases to predict charges and sentences, with text-classification models providing initial baselines. [38]. As the transformer architecture became dominant, models like BERT and T5 were increasingly used. CLC-UKET, released in 2024, compared fine-tuned BERT and T5 models to zero- and few-shot GPT-3.5 and GPT-5 to predict U.K. Employment Tribunal cases. [40].

2.2 Legal AI benchmarks and evaluation

The growth in frontier model capabilities has similarly coincided with increasingly ambitious benchmarks. LegalBench, released in 2023, is a major collaborative benchmark to which 40 authors contributed 162 tasks intended to assess legal domain expertise. The benchmark deliberately tests relatively clean legal propositions that can be reduced to a binary answer. [12].

Most recent benchmarks have sought to test higher-level judgment tasks, which usually lack an objective ground truth. JudgmentBench, released in May 2026, and Harvey's LAB targeted open-ended professional legal work, like drafting, analysis, and advice, for which there are potentially multiple defensible outputs and no independently verifiable ground truth [7, 41]. These benchmarks take different approaches to reward specification. JudgmentBench compared rubric-based scoring to comparative judgment (pairwise preference judgments collected from practicing attorneys), and found that comparative judgments recover the intended quality ordering substantially better than rubrics. LAB relies on a rubric-based approach, to allow rapid and scalable scoring of new model releases.

2.3 Verifiable rewards, forecasting, and outcome feedback

Frontier models are trained on several different forms of feedback, as described in prior work, including Reinforcement Learning with Verifiable Rewards (RLVR):. Recent work has used real-world (nonlegal) events to supply an objective training signal for RLVR. For example, Autocast, released in 2022, gathered forecasting questions from tournaments and a date-indexed news corpus, then had language models make predictions based on the information a human would have had at the time. [43]. Halawi and colleagues in 2024 used a retrieval-augmented forecasting system that could search for information, and generate and aggregate multiple forecasts [44]. ForecastBench, introduced in 2024, used this approach to create a dynamic benchmark that continually collects questions about unresolved future events and scores models once the event outcome is known [45].

Outcome-Based Reinforcement Learning to Predict the Future showed in 2025 that resolved binary forecasting questions could serve as reinforcement-learning rewards, training a 14-billion-parameter model on more than 100,000 events using adaptations of GRPO and ReMax [24]. OpenForecaster similarly explored training language models directly on resolved forecasting questions [25]. Some research in China has also used frontier models to predict criminal sentences [26].

3 LegalForecastBench: Dataset and evaluation protocol

3.1 Prediction target and information set

Let $c \in \{1, \dots, C\}$ index cases and $u \in \mathcal{U}_c$ index scored units within case c . Each unit identifies a challenged claim or a separately resolvable pleaded subclaim that a plaintiff has asserted against a moving defendant (or a group of similarly situated moving defendants). The "units" (i.e., the claims that each set of defendant(s) is moving to dismiss) are identified by labeling models that were given access to the motion papers.

When the benchmark is run, a model receives the same docket and motion papers that the judge received X_c and the specification of the units that are being challenged U_{cu} and returns

$$p_{cu} \in [0, 1], \quad p_{cu} \approx \Pr(Y_{cu} = 1 \mid X_c, U_{cu}), \quad (1)$$

where $Y_{cu} = 1$ means that the judge dismissed that specific claim-defendant unit in its entirety, in the target motion's first written disposition. That event is not necessarily a final victory in the lawsuit: a plaintiff could be given leave to amend the complaint (in which case the defendant typically will file a renewed motion to dismiss), or other claims might survive even if some are dismissed. The plaintiff also has a right to appeal final dismissal orders.

First written disposition of the frozen unit	Primary label
Entire claim dismissed, with or without leave to amend	1: fully dismissed
Only one nonseparable theory or remedy dismissed; claim remains	0: not fully dismissed
Claim dismissed against one defendant but survives against another	Separate labels for separately frozen units
Disposition does not reliably resolve the frozen target	Excluded from primary scoring

Table 1: Illustrative label semantics.

3.2 Cohort construction and labeling

The analyzed set consists of 91 U.S. federal cases with written decisions dated June 30–August 7, 2026. The cases, as labeled, contained 409 claim-defendant units that could be forecasted. Of those, 387 are scored. The unscored units involved situations where the plaintiff voluntarily withdrew the frozen claim, the order did not cleanly grant or deny the frozen claim, or (for four units) we determined after freezing the unit that the frozen defendant was the wrong target or the motion did not address the frozen claim. Of the scored claim-defendant units, 285 were fully dismissed and 102 were not. The sample was not random; our pipeline favored cases where some of the necessary documents could be obtained for free from online databases, although we typically still had to purchase a number of documents to obtain a complete set of the substantive motion papers that were before the judge.

The case construction and labeling pipeline used models as well as human legal oversight from me. The pipeline maintained separation between *unitization* and *outcome coding*. To avoid the possibility that the prediction units could leak information about the judge’s ultimate outcome to the models that were performing prediction, AI models first were asked to identify the challenged claim-defendant units based only on the the motion and the docket (without the decision). A later stage was given access to the judge’s ruling, to interpret the disposition of each frozen claim-defendant unit. I resolved disputed units (where two models did not agree on the disposition) and corrected labels as necessary. The process of creating the pipeline was iterative, and we did not track a blinded human-audit error estimate.

Models predicting outcomes were not given access to the judge’s ruling. The pipeline also attempted to screen dockets and filing text to avoid outcome leakage. Models had access to the motion to dismiss papers (through document text tools) in a sandbox with no Internet access. Model API calls were made with provider-side web search disabled. Provider-specific reasoning settings are retained in the run registries. Appendix A lists the configuration settings. The study population is the seventeen configurations in Table 2.

The decisions used in the benchmark were decided by the federal judges and became public, only after the reported knowledge or training cutoffs for all of the models that disclose such cutoffs. Kimi and Muse do not publish training cutoffs, but based on the model release dates, it appears unlikely that those models were exposed to the decisions in training.

The acquisition and labeling pipeline is designed to create new cohorts in the future, as new judicial decisions become available. This enables fresh comparisons even after models have been trained on the specific decisions at issue here. The downside of this approach is that refreshing the benchmark with a new cohort of cases changes the case mix and difficulty, so scores from different versions cannot be interpreted as a time series of model improvement.

3.3 Scoring and statistical inference

For $N = \sum_c |\mathcal{U}_c| = 387$ and $C = 91$, the primary metric is the micro-Brier loss

$$B_{\text{micro}} = \frac{1}{N} \sum_{c=1}^C \sum_{u \in \mathcal{U}_c} (p_{cu} - Y_{cu})^2. \quad (2)$$

The equal-case score changes the estimand by giving each case the same weight:

$$B_{\text{case}} = \frac{1}{C} \sum_{c=1}^C \frac{1}{|\mathcal{U}_c|} \sum_{u \in \mathcal{U}_c} (p_{cu} - Y_{cu})^2. \quad (3)$$

Lower is better for both. Micro-Brier treats each scored claim–defendant unit equally, so cases that involve a motion to dismiss many separate claims receive more weight than cases where only one claim and defendant was subject to a motion to dismiss. Equal-case Brier weights each prediction unit in inverse proportion to the number of units from that case.

The Brier rule is strictly proper for the binary target. If the true conditional event probability is q , then

$$\mathbb{E}[(p - Y)^2 \mid X, U] = (p - q)^2 + q(1 - q), \quad (4)$$

which is minimized by reporting $p = q$. The second term is irreducible uncertainty for that information set. A set of observable judicial events therefore create a verifiable score across the entire probability distribution, even though a single realized outcome standing alone does not reveal whether the probability a model assigned to that outcome was correct *ex ante* [18, 19]. Accuracy at threshold $p \geq 0.5$ is reported as a secondary descriptive metric.

The observed dismissal frequency is $\hat{\pi} = 285/387 = 0.73643$. Always predicting dismissal has accuracy 73.64%, so a constant forecast $p = \hat{\pi}$ has micro-Brier $\hat{\pi}(1 - \hat{\pi}) = 0.19410$. The constant is fitted retrospectively on this cohort, so it is a descriptive reference, not a baseline derived from earlier cases. The uninformative forecast $p = 0.5$ has Brier 0.25.

Log loss is a complementary diagnostic:

$$L = -\frac{1}{N} \sum_{c,u} [Y_{cu} \log p_{cu} + (1 - Y_{cu}) \log(1 - p_{cu})]. \quad (5)$$

For a wrong forecast assigning 0.99 to the predicted class, Brier loss is 0.9801 and log loss is approximately 4.605; at confidence 0.90 the corresponding losses are 0.81 and 2.303. Thus log loss imposes a large penalty on instances where the model assigned a very high probability to the outcome that did not occur.

Claim–defendant units from the same case share facts and legal issues; distinct claims in a single case may suffer from closely related legal defects, so the outcomes are not independent. We therefore measure within-case dependence using an observed-scale intraclass correlation coefficient (ICC) [72].

Across the 387 scored units in 91 cases, the observed-scale outcome ICC was 0.76 (95% CI: 0.61–0.87). Prediction losses also clustered within case because if a model failed to account, for example, for the possibility that a given motion might be denied on procedural grounds, all of the claims in a case that it expected to be dismissed could survive: GPT-6 Sol’s Brier-loss ICC was 0.42 (95% CI: 0.24–0.58). These intervals use 20,000 whole-case bootstrap resamples, and assume independence between cases.

4 Results

4.1 Frontier models substantially outperform unconditional references

Table 2 reports the seventeen configurations on the ranked leaderboard. GPT-6 Sol achieves the best observed micro-Brier (0.11763), equal-case Brier (0.12949), and accuracy (327/387, 84.50%). All seventeen beat the retrospectively fitted constant micro-Brier of 0.19410; GPT-6 Sol’s micro-Brier is lower than that constant by about 39.4%.

Sixteen configurations also beat majority-class accuracy (the 73.64% accuracy of predicting dismissal for every unit). Muse Spark 1.3 does not: 279/387 correct forecasts is 72.09%. Gemini 3.1 Pro

Configuration	Micro Brier	Case Brier	Correct	Acc. (%)	Cost (\$)
GPT-6 Sol	0.11763	0.12949	327/387	84.50	11.01
Claude Fable 5.1	0.12877	0.13653	318/387	82.17	83.18
Grok 4.6	0.13368	0.13480	312/387	80.62	15.72
Claude Opus 5	0.13987	0.13548	310/387	80.10	45.89
GPT-5.6 Sol	0.14009	0.13319	313/387	80.88	24.91
Kimi K3	0.14130	0.15367	303/387	78.29	16.87
GPT-5.6 Luna	0.14322	0.15084	313/387	80.88	1.48
Claude Opus 5.5	0.14540	0.14762	299/387	77.26	20.59
GPT-6.1 Sol	0.14948	0.13885	304/387	78.55	10.68
GPT-6 Astra	0.15221	0.14275	305/387	78.81	59.74
Gemini 3.8 Flash	0.15331	0.13900	310/387	80.10	35.61
Grok 4.7	0.15833	0.16454	298/387	77.00	33.00
Claude Sonnet 5	0.15848	0.15964	293/387	75.71	24.68
GPT-6 Luna	0.16368	0.17379	296/387	76.49	0.62
Claude Sonnet 5.5	0.17298	0.17183	295/387	76.23	6.99
Muse Spark 1.3	0.18121	0.17845	279/387	72.09	8.78
Gemini 3.1 Pro Preview	0.18345	0.18080	302/387	78.04	59.04
Cohort-rate reference	0.19410	—	285/387	73.64	—
Constant $p = 0.5$	0.25000	0.25000	—	—	—

Table 2: Benchmark results, sorted by micro-Brier loss.

Preview has the highest micro-Brier in the panel (0.18345) and still beats that accuracy baseline (302/387, 78.04%).

4.2 Confidence and calibration

Define confidence $h_{cu} = \max(p_{cu}, 1 - p_{cu})$ and the high-confidence subset $H_m = \{(c, u) : h_{cu} \geq 0.9\}$ for configuration m . Comparing the mean confidence in H_m with classification accuracy in H_m gives a useful conditional calibration diagnostic. Of course, this is not a full reliability curve; the subset differs by model, and its units remain clustered by case.

OpenAI models appeared more likely than others to make high-confidence forecasts. GPT-5.6 Sol, GPT-5.6 Luna, and Astra make high-confidence forecasts on 58.9%, 47.0%, and 53.5% of units respectively, compared with 20.9% for Fable 5.1. Their mean confidence in those subsets is approximately 95–97%, while realized accuracy is approximately 89–90%, suggesting some overconfidence, particularly for Astra, which missed 21 of 207 such predictions. Gemini 3.8 Flash also had relatively high sharpness, making 276 high-confidence forecasts (71.3% of units) and missing 35. Gemini 3.1 Pro Preview is sharper still, making 320 high-confidence forecasts and missing 57. GPT-6 Sol provides a useful counterexample within OpenAI, missing 8 of 169 high-confidence forecasts, for 95.3% subset accuracy. GPT-6.1 Sol is sharper and less accurate in that subset, missing 18 of 199. GPT-6 Luna misses 19 of 183. Claude Opus 5.5 misses none of 65.

Other high-performing models were less sharp. For example, Fable has no errors among 81 high-confidence units, while Kimi and Grok 4.6 each miss one. Claude Sonnet 5.5 makes 63 high-confidence forecasts and misses 5. The practical finding is that confidence varies substantially and it materially affects proper scores. Astra’s underperformance relative to cheaper models like Sol, for example, appears largely due to several high-confidence misses, which resulted in large Brier losses.

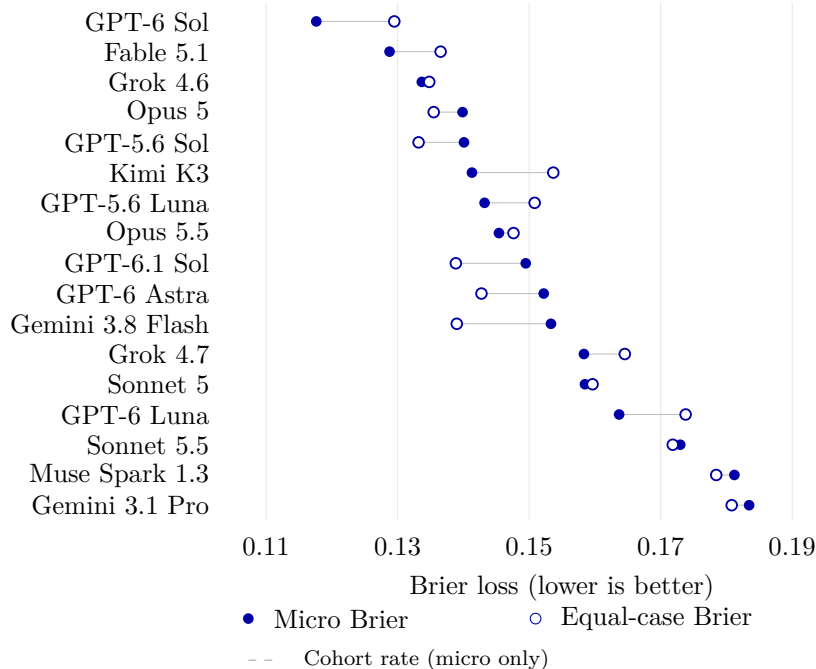


Figure 1: Micro- and equal-case Brier scores for the same seventeen configurations.

4.3 Inference cost

The normalized costs range from \$0.62 for GPT-6 Luna to \$83.18 for Fable. These are not actual invoice charges. For example, Fable was initially run without caching, resulting in an actual cost of \$189.70. The normalized price was reconstructed after the fact using turn-by-turn token counts, with the reconstruction validated against real billing data from other Anthropic model runs. In several other cases, provider discounts that we received were normalized to standard API rates. On these standardized estimates, GPT-6 Luna, GPT-5.6 Luna, and GPT-6 Sol form the observed micro-Brier cost–score frontier.

4.4 Evaluating Jev, a non-autoregressive discriminative decision model

Jev—described by its developer as a probabilistic “System One” model—was evaluated as a separate experiment because its context window and lack of agentic tool calling differ from the document-enabled configurations used with frontier models. [49]. GPT-5.6 Luna summarized the motion documents, without commentary or analysis. Jev received the summaries, and the unit questions, in a single request per case, without document tools or multiple turns. In a second condition, Grok 4.6 prepared a separate set of shorter summaries of the same 482 documents, which Jev received under the same protocol.

Jev was not well-calibrated for this task. Jev’s micro-Brier of 0.36754 is worse than the 0.25 loss of always reporting $p = 0.5$. Its mean predicted dismissal probability was approximately 0.305, while the observed frequency in this data set was 0.736. This resulted in severe underprediction in the aggregate. Its 32.6% threshold accuracy is also below majority-class prediction.

We tested Luna using the same summaries that Jev received in order to determine whether Jev’s underperformance was attributable to receiving only summaries of the litigation documents, rather than the full documents used in our primary evaluation. Luna performed much better than Jev,

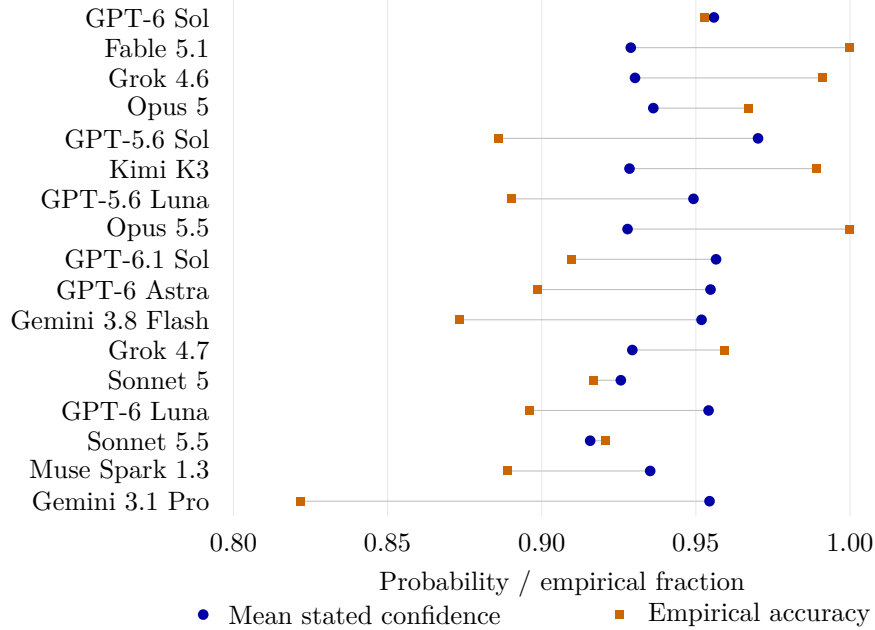


Figure 2: Conditional confidence diagnostic on predictions with $\max(p, 1 - p) \geq 0.9$. Each row uses the subset selected by that model, rather than a common set of units. Wrong/selected counts: GPT-6 Sol: 8/169; Fable 5.1: 0/81; Grok 4.6: 1/112; Opus 5: 4/121; GPT-5.6 Sol: 26/228; Kimi K3: 1/92; GPT-5.6 Luna: 20/182; Opus 5.5: 0/65; GPT-6.1 Sol: 18/199; GPT-6 Astra: 21/207; Gemini 3.8 Flash: 35/276; Grok 4.7: 4/98; Sonnet 5: 5/60; GPT-6 Luna: 19/183; Sonnet 5.5: 5/63; Muse Spark 1.3: 15/135; Gemini 3.1 Pro: 57/320. This is not a full reliability diagram or an estimate of statistical significance.

even when it only had access to the same summary Jev received. GPT-5.6 Luna with the identical summaries achieves micro-Brier 0.15489 and accuracy 78.6%, demonstrating that the summaries contained useful predictive signal that Jev was unable to exploit (although Luna performed better in the primary evaluation, where it had full document access, which could mean that the summaries lost some relevant signal or that the interactive environment allows for better performance).

We also tested whether a different summary pipeline would change Jev’s result. With the Grok summaries, Jev improved to micro-Brier 0.32725, equal-case Brier 0.30505, and 38.2% accuracy; the improvement over Jev with Luna summaries is significant on all three metrics (paired case-cluster bootstrap, Bonferroni-corrected across the six pairs of the four summary conditions and three metrics). Jev nevertheless remained significantly worse than both Luna summary controls and worse than the constant $p = 0.5$ forecast; its mean predicted dismissal probability rose only to about 0.356. Because Grok wrote different summaries and no Luna control received them, this compares

Condition	Micro Brier	Case Brier	Acc. (%)
GPT-5.6 Luna full documents	0.14322	0.15084	80.88
GPT-5.6 Luna summary + high reasoning	0.15489	0.16119	78.55
GPT-6 Luna summary + no reasoning	0.17152	0.18694	74.16
Jev Luna summary + one-shot	0.36754	0.34416	32.56
Jev Grok summary + one-shot	0.32725	0.30505	38.24

Table 3: Results from the Jev experiment, with Luna evaluating the same Luna summaries as a control. No control received the Grok summaries.



Figure 3: Probability quality with a shared summary cache. Both Luna conditions exploit predictive signal in the summaries, while Jev scores worse than the constant $p = 0.5$ forecast with either set of summaries. The full-document reference changes both information access and interaction. Connecting segments indicate weighting sensitivity, not uncertainty intervals.

pipelines; it does not isolate summary quality or model ability.

Summary preparation cost an estimated \$10.32 for the Luna summaries and \$29.99 for the Grok summaries, both reconstructed from recorded usage. An earlier Grok preparation under the standard summary prompt cost a further \$2.00, and \$4.11 in uncertain Grok preparation charges is held separately and excluded from these totals. Successful inference estimates are approximately \$0.044 for Jev with Luna summaries, \$0.020 for Jev with Grok summaries (provider-reported), \$1.62 for GPT-5.6 Luna (with high reasoning), and \$0.103 for GPT-6 Luna (which was run without reasoning as a control).

5 Potential advantages of real-world litigation data over synthetic data or rubric-based rewards

5.1 Evaluating open-ended legal work

JudgmentBench compares two ways of evaluating open-ended legal work: asking practicing attorneys which of two outputs they prefer, and asking them to score outputs against specified criteria. Across 30 legal tasks with outputs at three constructed quality levels, comparative judgments recovered the intended quality ordering substantially better than rubric scores and required less than half the annotation time [7]. The constructed quality levels and task selection limit what this establishes about professional work more broadly. Together with judicial outcomes, these approaches provide three distinct evaluation signals: what a court did, which work product an expert prefers, and whether an output satisfies specified requirements.

Preferences and rubrics can assess aspects of advice, analysis, and drafting that a forecast score does not directly measure. Rubrics make the requirements explicit; comparative judgments let reviewers weigh the work as a whole without an exhaustive checklist. Either can supply evaluation data, and either can be used to construct training rewards. Collecting preference judgments for evaluation is not itself reinforcement learning.

Harvey’s Legal Agent Benchmark (LAB) provides a large-scale case study of the rubric approach. Launched on May 6, 2026, it contains 1,200 tasks spanning 24 practice areas and more than 75,000

criteria, with LLM judges scoring model outputs [41, 57]. The LAB-AA partnership with Artificial Analysis extends its use in evaluation [58].¹ Harvey and Applied Compute also report full-parameter reinforcement learning of a legal agent using rubric-based rewards [9]. Harvey’s August 2026 Tenet research preview reports gains on LAB and other legal benchmarks after training a model based on Kimi K3 [10]. These are company-reported training results; they make the quality of the criteria consequential for learning as well as evaluation.

5.2 Audit of LAB’s litigation rubrics

A rubric can assess richer work than a judicial forecast, but that broader coverage is useful only if the rubric specifies the task correctly and the grader applies it reliably. Errors can arise in the requested work, the supplied record, the legal answer prescribed by a criterion, or the grader’s treatment of a model response. These are distinct failure modes. The underlying question is one of construct validity: whether the score measures the capability the benchmark claims to assess [30].

Our retained AI-assisted review covers the pinned litigation-and-dispute-resolution subset: 52 tasks and 2,858 criteria. GPT-6 Sol flagged 616 criteria; Claude Opus 5.5’s final reconciliation flagged 564. Their union contains 835 flagged criteria, leaving 2,023 unflagged by either final report. Of 345 overlapping flags, both reviewers called 83 problematic and 135 arguable; 127 received different levels of concern. The union represents 29.2% of the 2,858 criteria; the other 70.8% were not flagged, which does not establish that they are correct. These are review outputs, not a verified error rate. Opus’s reconciliation saw Sol’s findings, so agreement in the final reports is not independent confirmation [27].

To measure how many flags reflect genuine defects, the author, a practicing litigator, reviewed a simple random sample of 25 of the 835 flagged criteria, drawn with a fixed seed from 19 tasks (Appendix C). A criterion counted as defective if, applied as written to the task as given, it would fail a response a careful litigator would consider correct or pass one that is wrong. 16 of the 25 were defective in ways that affect grading, and 9 of those were unambiguous objective errors in which the supplied record, the governing law, the cited authority, or the arithmetic contradicts the rubric. Another 3 were arguable and 6 were not defective. Extrapolating to the flagged population, at least 383 of the 835 flagged criteria are defective (one-sided 95% bound; point estimate 534), or more than 13.4% of all 2,858 criteria in the practice area; at least 171 are unambiguous objective errors. Agreement between the two AI auditors was informative: 9 of the 10 sampled criteria flagged by both were defective, against 7 of the 15 flagged by only one.

The defects were not confined to rubric text. 14 of the 25 sampled criteria come from task environments that themselves contain unplanted errors, such as invoices whose time entries do not sum to the stated fees or production indexes whose counts contradict one another; these span 10 of the 19 sampled tasks, and only 4 of the 25 criteria involved neither a defective criterion nor a defective environment. Extrapolated, at least 319 flagged criteria (more than 11.1% of all criteria) come from such environments. A separate AI-assisted check of every case citation in the 19 sampled environments against CourtListener, with spot checks, found problems in 45 of 102 citations to purported real authority: 32 real cases cited for holdings, quotations, or facts they do not contain, 7 cases that could not be found, and 5 garbled citations to real cases [28]. These estimates cover only flagged criteria, rest on a single reviewer’s judgment, and count criteria rather than independent

¹Harvey’s product announcements report LAB all-pass results for Opus 4.8 (10.4%, May 28), Fable 5 (13.3%, June 9), and Sonnet 5 (5.8%, June 30) [61–63]. These are Harvey’s evaluations of third-party models, not evidence that each developer independently adopted LAB as its definitive legal benchmark. Prominence does not establish evaluation validity.

mistakes.

Errors of this size are large relative to the margins that separate models on LAB. On Artificial Analysis’s Harvey LAB-AA leaderboard, Muse Spark 1.3 leads with a criterion pass rate of 95.5%, followed by Kimi K3 at 94.6%, Claude Fable 5.1 at 93.3%, Claude Opus 5.5 at 91.2%, and GPT-5.6 Luna at 87.9%; each model is run once and no confidence intervals are reported [59]. Harvey’s own announcements separate models by a few points of all-pass rate, such as Fable 5 at 13.3% against Opus 4.8 at 10.4% [62]. LAB-AA scores a private 120-task set spanning all 24 practice areas, not the public litigation tasks we audited, and its implementation differs from Harvey’s [60], so our estimate does not transfer directly. But if criteria elsewhere in LAB are defective at anything like the rate we found, more than 13.4% of the criteria in this practice area, rubric error is as large as or larger than the reported gaps between leading models, and it is unclear how much of those gaps it could explain. All-pass scoring compounds the problem, because a single defective criterion fails an otherwise acceptable task.

The defects also did not fall evenly on the two models we ran through LAB’s own grading pipeline, GPT-6 Luna at extra-high reasoning effort and Claude Opus 5.5 at low effort, each graded by LAB’s two judges (Claude Sonnet 4.6 and GPT-5.5). Of the 16 sampled criteria the author found defective, 8 were graded differently for the two models. In 2, both judges failed a defensible Luna answer while passing Opus, which had added material the rubric demanded but the task did not need; in a third, one judge did the same. In 1, the pattern ran the other way: both judges failed Opus for the better answer, while one judge passed Luna for repeating the rubric’s incorrect count. The remaining splits reflect judge inconsistency or Luna errors unrelated to the defect, and in 3 other defective criteria both models were failed on legitimate work. Luna failed more criteria overall, the two runs used different reasoning settings, and none of these splits changed a task’s all-pass result, so the sample does not show that the defects systematically favor one model. It does show that a defective criterion can reorder models, rather than only adding noise that affects every model alike.

If a defective criterion supplies a training reward, it also creates a problem of reward misspecification. Reward hacking is a possible learned response; reward-model overoptimization is a possible divergence between an improving proxy score and the underlying target [31, 32]. Our audit examines tasks and criteria, not a training trajectory. It therefore cannot establish that rubric-based RL is ineffective, that synthetic tasks generally harm performance, or that training on litigation tasks caused degradation.

OpenAI’s 2026 analysis of SWE-bench Verified provides a useful methodological analogy: its selected audit of 138 difficult tasks reported problems involving test requirements and task specification [48]. That finding supports examining alignment between tasks and their scoring rules; it provides no defect estimate for LAB. A stronger legal study would compare matched environments built from synthetic and authentic records, independently adjudicate their criteria, and evaluate transfer to held-out professional tasks.

5.3 A motion-to-dismiss case study

The task `analyze-counterparty-motion-to-dismiss` requests an opposition issues memorandum in a fictional software-services dispute involving alleged pre-contractual representations, implementation defects, and jurisdictional, venue, and pleading defenses. The AI reviewers flagged 16 of its 34 criteria. My targeted review uses the task and supplied documents at the pinned upstream revision recorded by the local audit [29]. The analysis below distinguishes direct record observations from legal interpretations and unverified AI flags; it does not treat every flag or solver failure as a

grading defect.

Forum selection and litigation strategy. Complaint paragraph 48 sets out a clause selecting state or federal courts in Travis County, Texas, while the action is filed in the Northern District of Georgia. Paragraphs 18–19 assert statutory venue in Georgia, and paragraph 123 argues that the statutory deceptive-practices claim falls outside the clause. Those arguments may be weak, but the complaint is not wholly silent about the forum issue. A rigorous task should ask whether transfer is likely, which claims the clause covers, and whether a procedural objection changes the client’s position. Under *Atlantic Marine*, a valid clause selecting another federal forum is generally enforced through section 1404(a), while a state or foreign forum implicates forum non conveniens [64]. Correcting an opponent’s procedural vehicle is not the same as establishing that the chosen Georgia forum will retain the dispute. A criterion that emphasizes the former without measuring the latter can reward issue identification without strategic assessment.

Accrual and pleaded dates. Criteria C-013–C-015 require an accrual dispute and alternative discovery dates. The complaint alleges discovery of integration problems in August 2022, latency problems by November 2022, an acknowledgment in January 2023, and an unsuccessful patch on April 15, 2023. Paragraph 83 alleges that the plaintiff could not know the full extent of the breach until the patch failed, claims reliance on a promised fix, and states a March 22, 2024 filing date. These allegations require analyzing knowledge sufficient for accrual, distinct breaches, tolling or estoppel, and enforceability of the one-year contract term. Texas law itself restricts contractual limitations periods, subject to statutory exceptions [65].

The appropriate critique is therefore not that the dates mechanically prove the entire claim time-barred. Nor is it enough to announce that knowledge is always a fact issue immune from dismissal. A plaintiff’s allegations can establish an affirmative defense on the face of the pleading, while a genuine factual or legal issue may prevent that result [66]. The rubric should require applying the governing rule to the actual chronology and contract. It should allow the model to distinguish stronger from weaker opposition arguments instead of requiring the conclusion that every accrual question is unresolved.

Integration versus non-reliance. Criterion C-012 requires the model to characterize the merger-clause defense as legally incorrect. MSA section 14.7 also contains express non-reliance language, which is analytically distinct from an ordinary integration clause. Texas authority distinguishes these contractual concepts and examines the language and surrounding circumstances [67]. The source record therefore does not justify a categorical rubric requiring the ordinary-merger-clause answer without that analysis. This observation identifies an overbroad criterion; it does not establish that the particular disclaimer is enforceable or that the fraud claim must fail.

Hidden output requirements. C-028 requires concrete opposition recommendations for at least eight issues, although the prompt supplies no minimum count. C-022–C-025 prescribe high-severity treatment of specified issues. These criteria permit equivalent severity language; it would be inaccurate to say they necessarily require literal labels. C-026 prescribes organization tied to the motion’s framework and expressly permits omitting Rule 12(b)(1) when no genuine issue exists. Whether those requirements improve usefulness to counsel is a measurement question. They should not be confused with independently established legal errors.

Document construction and limits of the review. The supplied documents have programmatic-generation metadata and include a stale table-of-contents instruction in the agreement. The motion nevertheless contains a caption, table of contents, and table of authorities. We have not established a court-rule violation warranting striking it, proven LLM authorship, or estimated how often such artifacts occur across LAB. Synthetic design can be useful for controlled testing; the concern is whether the rewarded conduct transfers to actual matters. The local audit also records genuine solver mistakes and unresolved legal disagreements. A benchmark-wide conclusion would require a representative expert review of source documents, criteria, model outputs, and grading decisions.

5.4 Comparing supervision signals

Judicial outcomes. Outcomes provide an external reference that the evaluator does not choose. The task designer must still define the event, separate pre-decision information from later evidence, and interpret the disposition correctly. Once that target is fixed, a proper scoring rule evaluates the forecast against what happened without requiring an expert to prescribe the best legal argument. The coverage is narrow: predicting a disposition does not directly evaluate the quality of a brief, the usefulness of advice, or the faithfulness of an explanation. A successful forecast may combine doctrinal analysis with institutional priors rather than identify the legally correct answer.

Rubrics. Rubrics can directly assess several dimensions of professional work, including factual support, legal analysis, completeness, and compliance with a client’s instructions. Their difficulty is specification: the criteria must distinguish indispensable analysis from optional presentation and allow defensible alternative answers. All-pass aggregation magnifies this problem because one defective criterion can make an otherwise useful work product fail. Some requirements are indispensable, so all-pass scoring is not inherently inappropriate; the question is whether the required items are valid. Nor does learning from rubric rewards necessarily produce only rubric compliance: Gunjal and colleagues report gains on the multiple-choice GPQA-Diamond benchmark relative to direct Likert-based judge rewards, although that finding does not establish transfer to legal work [8].

Preferences. Comparative judgments can capture professional assessments that are difficult to specify in advance, including whether a memorandum prioritizes the issues that matter. They also depend on reviewer expertise, objectives, and consistency. Disagreement may reflect different defensible judgments or measurement noise, and a preferred output need not be the most accurate or useful one. Research outside law shows that human preferences can favor persuasive agreement with a user’s beliefs over factual correctness, and that optimizing a learned reward can amplify such biases [33, 34]. In legal evaluation, a corresponding concern is that a reviewer might prefer confident or congenial advice to a better-supported adverse assessment. This is a risk to investigate, not a finding that JudgmentBench’s attorney judgments exhibit sycophancy. Preference evaluation and preference-based training must be assessed separately.

These signals are complementary. Authentic case records can support an outcome forecast, a rubric checking necessary legal and factual requirements, and a blinded expert comparison of the resulting advice or drafting. Reporting these assessments separately would make disagreements visible: a model might forecast accurately while giving poor advice, or produce polished work that misses the issue most likely to determine the result. The research question is which signals, individually or together, support useful legal judgment beyond the tasks on which they are scored.

6 Future work: from outcome evaluation to outcome feedback

6.1 Outcome feedback as a training signal

LegalForecastGym implements a separate, single-step forecasting environment intended for post-training experiments. It renders a static litigation packet and receives a structured action containing a dismissal probability and cited evidence spans. Training loaders admit only explicitly designated training collections and reject locked test matters. Episode assembly keeps the label out of the observation. The current phase does not implement a multi-turn litigation simulator or train an agent to take procedural actions.

The checked Gym artifacts contain no completed parameter-update experiment or held-out before/after training comparison. They do contain a live sampling comparison on 50 synthetic development episodes, used to assess output format and select an open-weight sampling model. That run is infrastructure evidence, not evidence of legal forecasting improvement. A separately recorded Gym export of 100 matters and 425 scoreable units is also distinct from the 91-case, 387-unit benchmark analyzed here; those populations are not pooled.

The intended learning study asks whether earlier outcome feedback improves predictions on later, disjoint matters and whether any gains extend beyond forecasting. Parameter learning, in-context feedback, and changes to the surrounding agent system are different interventions. Each needs a matched control and a held-out endpoint. Forecasting literature makes such learning plausible, and OpenForecaster reports calibration transfer to other question-answering benchmarks; those findings do not establish improved litigation strategy [24, 25]. Appendix D specifies a concrete prospective design.

6.2 Testing transfer beyond forecasting

A reduction in forecasting loss on later cases would be evidence of improvement on the prediction task. Transfer would require improvement on additional legal tasks assessed independently of that score. The prospective study in Appendix D would therefore compare the trained model with a matched control on issue prioritization, strategic advice, and drafting using later, disjoint matters. The relevant questions are whether the model selects the strongest arguments, relates its advice to the client’s objectives and litigation risks, and produces work supported by the record and applicable law. Explanations should be evaluated separately from the probabilities they accompany.

Practicing lawyers would assess paired outputs with model identity and training condition concealed and presentation order randomized. Both models would receive the same pre-decision record and client instructions; reviewers would initially assess the work without seeing the eventual disposition to limit hindsight effects. Expert comparisons would be supplemented by checks of factual and legal support, with reviewer disagreement reported. Improvement on these independent assessments would provide evidence of transfer; a better forecast score alone would not. LegalForecastGym provides infrastructure for testing this hypothesis, not a completed learning or transfer result.

7 Limitations

Selection into adjudication. The target population is a set of accessible cases with written motion decisions, not all disputes, all filed cases, or all contemplated motions. Parties settle, abandon claims, amend pleadings, or choose not to sue. Cases reaching a published disposition may differ in uncertainty, stakes, representation, and bargaining conditions. Calibration conditional

on reaching this dataset does not establish calibration for counterfactual outcomes in the broader dispute population [52]. At the same time, selection does not prove that learned skills cannot transfer: that is an empirical question requiring an independently defined evaluation population.

Judge, court, and party priors. We have not established how much performance comes from close legal analysis rather than judge, court, party, or case-type priors.

Labels and completeness. Unitization can omit, merge, or split legally distinct targets; dispositions can be ambiguous or procedural. Human oversight helps but does not quantify residual error. The absence of a complete blinded label audit and acquisition/exclusion census limits claims about both label accuracy and representativeness. Excluding 22 unscored units preserves an interpretable target but may also change difficulty. These limitations deserve the same scrutiny as disputed rubrics in another benchmark.

Statistical and execution limits. One successful prediction per unit does not characterize run-to-run variance, operational reliability, or expected total cost. The cohort is small and imbalanced. Case clustering does not resolve unverified dependence among related cases. Reported scores compare model-harness-provider configurations, not isolated model weights.

Temporal and transfer limits. A provider-reported cutoff and disabled browsing reduce specific leakage risks without proving contamination absent. New benchmark versions change the case distribution. Generalization to other jurisdictions, procedural stages, settlement decisions, drafting, or strategic advice is not established by this study.

8 Conclusion

LegalForecastBench operationalizes an uncertain but consequential legal judgment as a probabilistic forecast of a later judicial event. In the beta, all seventeen document-enabled configurations outperform a cohort-fitted constant in micro-Brier, while confidence behavior and model ordering differ in informative ways. GPT-6 Sol has the strongest observed scores, and the shared-summary comparison exposes a large gap between Jev and Luna that cannot be explained by the summaries containing no signal.

Real outcomes provide an external scoring reference, but useful training signals still require careful task definition, information boundaries, and legal labeling. The next research question is whether feedback on such forecasts improves performance on later cases and transfers to independently assessed legal work. The present evidence motivates that question; it does not answer it.

Reproducibility, access, and disclosure

This manuscript reports the seventeen-configuration comparison and checks its numerical claims against the retained repository artifacts as of September 30, 2026. The accompanying aggregate data and figure script regenerate the plots without court-document bytes. Code and result artifacts are maintained in the [LegalForecastBench repository](#); access to restricted source documents and private adjudication material is a separate audit process. Availability of code is not equivalent to unrestricted source-data replication.

The author designed the benchmark and supplied domain judgments. AI assistants supported this manuscript’s drafting, literature retrieval, and numerical and source checks. The LAB audit combines an AI-assisted review of every criterion in one practice area with the author’s review of a random sample of the flagged criteria; no second human reviewer has re-decided that sample. No model-weight reinforcement-learning experiment is claimed unless explicitly identified as a result of an external cited study.

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A Model configuration and temporal evidence

The table records the maintained local evidence, rather than certifying provider training histories. “K” denotes a reported knowledge cutoff, “T” a reported training cutoff, and “?” no established cutoff. Month-only dates are compared as the first day of the reported month. The reported cutoffs for fifteen configurations precede June 30, 2026. Kimi and Muse remain qualified. A release date is not used to fill an unknown cutoff.

Configuration	Reasoning	Reported cutoff	Relation to window
GPT-6 Sol	High	2026-04-20 (K)	Precedes window
Claude Fable 5.1	Adaptive/default	2026-06 (T)	Precedes window
Grok 4.6	High	2026-02-01 (K)	Precedes window
Claude Opus 5	Adaptive/default	2026-05 (T)	Precedes window
GPT-5.6 Sol	High	2026-02-16 (K)	Precedes window
Kimi K3	High	Unknown (?)	Unknown
GPT-5.6 Luna	High	2026-02-16 (K)	Precedes window
Claude Opus 5.5	High	2026-06 (T)	Precedes window
GPT-6.1 Sol	High	2026-04-30 (K)	Precedes window
GPT-6 Astra	High	2026-04-30 (K)	Precedes window
Gemini 3.8 Flash	High thinking	2026-03 (K)	Precedes window
Grok 4.7	High	2026-05 (K)	Precedes window
Claude Sonnet 5	Adaptive/default	2026-01 (T)	Precedes window
GPT-6 Luna	High	2026-05-18 (K)	Precedes window
Claude Sonnet 5.5	High	2026-06 (T)	Precedes window
Muse Spark 1.3	High	Unknown (?)	Unknown
Gemini 3.1 Pro Preview	High thinking	2025-01 (K)	Precedes window

Table 4: Configuration and cutoff evidence retained by the benchmark. “Precedes window” is a date comparison under the maintained rule, not an independently verified training-history claim.

The common document interface standardizes access, not every inference setting. Claude Fable 5.1, Claude Opus 5, and Claude Sonnet 5 use adaptive thinking at provider defaults. Claude Opus 5.5 and Claude Sonnet 5.5 pin high effort. The other configurations use the high reasoning or thinking setting recorded in their registries. Provider-default sampling and serving updates may still introduce variation. Full-document predictions and the one-request shared-summary conditions are kept separate throughout the analysis.

B Result provenance

The primary cohort identifier is `cycle-1-91-2026-09-08-luna-r5`. Costs in the main table are the totals recorded on the run receipts from provider-reported token usage and the frozen rates.

The checked source revisions are [LegalForecastBench 58371a0e](#), [LegalForecastCorpus b392ddaf](#), and [LegalForecastGym 4cc0f528](#). The LAB source snapshot is [1dd81403](#). These identify inspected versions; they do not certify every execution against the current source tree.

C Human review of AI-flagged LAB criteria

After noticing instances in which LAB tasks appeared to contain errors, I asked two frontier models, GPT-6 Sol and Claude Opus 5.5, to audit the 52 tasks in LAB’s litigation-and-dispute-resolution practice area at Harvey commit 1dd81403. The goal was to identify rubric criteria that misstate the law, and environments or task specifications containing errors or deficiencies that would make the tasks unreliable for benchmarking or for training models. Harvey has used LAB for both [9, 10]; [confirm the source for public tasks used in training and a held-out set used for benchmarking](#).

The models flagged 835 of the 2,858 criteria (29.2%) as problematic or arguable. GPT-6 Sol flagged 616. Claude Opus 5.5, working without access to Sol’s results, flagged 650; it then reviewed every flag from both passes against the record and the governing law, ending with 564. The 835 figure is the union of Sol’s flags and Opus’s final flags [27].

To establish how many of these flags reflect genuine defects, I reviewed a simple random sample of 25 of the 835 flagged criteria, drawn with a fixed seed under a protocol committed before the review began [28]. The sample spans 19 tasks. For each criterion I examined the task instructions, the supplied documents, the governing law, both auditors’ findings, and how LAB’s own judges graded two model runs on it. A criterion is defective when, applied as written to the task as given, it would grade a competent response wrong: failing a response a careful litigator would consider correct, or passing one that is wrong. Two applications of that standard were made explicit during the review. A criterion that requires one of several permissible approaches is defective when it rejects an approach a careful litigator would adopt on the record. And an unplanted error in the supplied record, such as figures that do not add up or misdescribed case law, makes a criterion defective when it changes what a competent response can get right. Separately, I recorded whether each task environment itself contains a defect, whether or not that defect affects the sampled criterion.

Of the 25 criteria, I found 16 defective in ways that affect grading, 9 of them unambiguous objective errors in which the record, the law, the cited authority, or the arithmetic contradicts the rubric. 3 were arguable and 6 were not defective. 14 of the 25 come from task environments that contain defects (10 of the 19 tasks). Only 4 involved neither a defective criterion nor a defective environment (Table 5).

Treating the sample as a draw without replacement from the 835 flagged criteria, the exact one-sided 95% lower confidence bounds are: at least 383 flagged criteria are defective in ways that affect grading (point estimate 534; the bound is 13.4% of all criteria in the practice area); at least 171 are unambiguous objective errors (point estimate 301); and at least 319 come from task environments that contain defects (point estimate 468). At least 524 involve a defective criterion, a defective environment, or both (point estimate 668). In the sample, 9 of the 10 criteria flagged by both auditors were defective, against 7 of the 15 flagged by only one.

These estimates cover only criteria the auditors flagged; defects among unflagged criteria are not estimated. They count criteria, not independent mistakes: one error in a task environment can make several criteria defective. Each verdict is my professional judgment as a litigator, made after an AI stress test that argued both for and against each call, and no second human reviewer has re-decided the sample. A separate AI-assisted check of every case citation in the 19 sampled task environments against CourtListener, with spot checks, found problems in 45 of 102 citations to real authority, including 7 cases that could not be found and 32 real cases cited for propositions they do not contain [28].

Criterion verdict	Environment defective	Environment correct	Total
Defective	10	6	16
Arguable	2	1	3
Not defective	2	4	6
Total	14	11	25

Table 5: Reviewer verdicts on the 25 sampled criteria, by whether the task environment itself contains a defect.

The sampled criteria

Each entry gives the criterion as written, the reviewer’s verdicts, how LAB’s two native judges (Claude Sonnet 4.6 and GPT-5.5) graded the GPT-6 Luna and Claude Opus 5.5 runs on that criterion, and the reviewer’s analysis. Harvey task files are cited at commit 1dd81403.

1. Assess Reasonableness of Staffing Levels on Litigation Invoice (C-049)

Criterion. **Calculates budget overage percentage as approximately 17%.** PASS if the memo calculates the percentage by which the attorney fee subtotal (\$193,372.50) exceeds the approved budget ceiling (\$165,000) as approximately 17% (exact figure is 17.2%). FAIL if no percentage overage calculation is provided or if the percentage stated is materially incorrect (off by more than 2 percentage points).

Verdict. Defective (unambiguous objective error, deficiency in task environment). *Environment.* Defective: the invoice’s 117 time entries add up to \$215,680.00, while its summary and subtotal show \$193,372.50. *AI auditors’ reasoning (reviewer’s assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: F/F; Claude Opus 5.5: P/P.

Analysis. The rubric is testing the AI on a specific percentage calculation from an invoice, but the document in the environment is internally inconsistent. The invoice’s summary page and its subtotal row show fees of \$193,372.50 (17.2% over the \$165,000 budget), but the 117 detailed time entries add up to \$215,680.00, which puts the overage at \$50,680, or 30.7% (13.5 points from the rubric’s required 17.2% answer and far outside the 2-point tolerance).

2. Categorize Document Production Set by Relevance and Privilege — Privilege Log and Relevance Classification Report (C-012)

Criterion. **ISSUE_003: DOC_012 identified as Partially Privileged.** PASS if DOC_012 (chandrasekaran-meeting-notes-aug28.docx, Chandrasekaran’s handwritten meeting notes from August 28, 2023) is classified as Partially Privileged. FAIL if classified as fully Privileged or Not Privileged.

Verdict. Defective (low-quality professional judgment, rubric is incorrect). *Environment.* Correct. *AI auditors’ reasoning (reviewer’s assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/P; Claude Opus 5.5: P/P.

Analysis. The rubric is wrong because it fails fully Privileged, which is the better answer here (even if Partially Privileged is defensible).

- The documents are handwritten lawyer notes from a fictitious meeting discussing problems with the R&D Department. The headings on individual topics seem intended to create the

impression that there is a discussion of business topics (which would not be privileged) followed by a separate discussion of legal topics (privileged), so from a superficial glance, the rubric does seem to match the topic headings. However, the bullets under each heading make clear that the entire meeting is just dedicated to the topic of the company’s desire to find a way to fire the head of R&D without getting sued. The business topics (which look to be nonprivileged) are discussions of how the R&D department is a bit of a disaster, such that management wants to fire the head. The legal sections then involve discussion of the fact that the R&D head has some sort of environmental whistleblower status, so termination is likely to trigger a lawsuit/whistleblower retaliation claim.

- The meeting was specifically convened because outside counsel was concerned about the retaliation claim, and even some of the ‘ops’ sections that are supposed to be nonprivileged reflect counsel’s advice, so they cannot be cleanly treated as nonlegal business discussions. For example, the ‘Ops’ sections record HR “compiling the file” of performance documentation and lay out a “Performance improvement plan first” / “Documentation period” sequence, which carries out outside counsel’s recommendation, recorded in the legal section, to build a “contemporaneous performance record to insulate against retaliation” claims.
- While partially privileged might be defensible, my preferred answer would be fully privileged, which the rubric incorrectly fails.

3. Compare Document Production Against Discovery Requests — Discovery Gap Analysis Memorandum (C-029)

Criterion. **ISSUE_011: Notes only 5 documents actually relate to counterclaim damages.** PASS if the memorandum notes that only 5 of the 14 documents coded to RFP 27 (MLC-004056 through MLC-004102) actually relate to Pacific Corridor’s volume shortfall and the counterclaim damages calculation, meaning the counterclaim damages support is thin. FAIL if this is not noted.

Verdict. Defective (unambiguous objective error, rubric is incorrect). *Environment.* Defective: the production index codes 15 rows to RFP 27, but its summary tab gives a count of 14 for a range that contains 13. *AI auditors’ reasoning (reviewer’s assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: F/P; Claude Opus 5.5: F/F.

Analysis. This task requires models to make the point that “only” five documents in the bundle “actually relate to Pacific Corridor’s volume shortfall and the counterclaim damages calculation, meaning the counterclaim damages support is thin.” This rubric is flawed on multiple levels.

- First, it overlooks MLC-003781, a sixth document that plainly relates to the volume shortfall, so the count is objectively wrong (and the agents pointed to several other examples, at least some of which also seem to fit within the stated criteria). There is a document in the bundle that claims that only five documents concern the volume shortfall, but that assertion is undercut by other evidence supplied in the environment.
- Second, this rubric is someone’s argumentative characterization of the documents and probably would not hold up under adversarial testing. For example, there is case law that the term “relate to” is fairly broad and flexible, so quite plausibly a number of other documents would apply.
- Third, the premise underlying this rubric—that “only” five documents is such “thin” evidence that it must be called out—is wrong. Even one document could constitute conclusive proof of damages (the general rule is that damages only need to be reasonably estimated; courts do

not require great precision); to my mind, this is actually quite a lot of evidence at the fact stage compared to what one would normally see.

4. Draft Case Assessment Memorandum — Litigation Risk Analysis for Distribution Agreement Dispute (C-025)

Criterion. **ISSUE_007: Recommends motion to dismiss on unjust enrichment.** PASS if the memo recommends filing a motion to dismiss (or motion for judgment on the pleadings, or similar dispositive motion) targeting the unjust enrichment claim (Count IV) on the ground that an express contract governs the parties' relationship. FAIL if no such motion recommendation is made regarding the unjust enrichment claim.

Verdict. Defective (low-quality professional judgment, rubric is incorrect). *Environment.* Correct. *AI auditors' reasoning (reviewer's assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: F/F; Claude Opus 5.5: P/P.

Analysis. A different rubric on this task concludes that plaintiff has a strong breach of contract claim. (See id: C-006.) The prompt instructs models to review case documents and “prepare a comprehensive case assessment memo,” and models are dinged on this rubric if they do not “recommend[] filing a motion to dismiss” on the unjust enrichment claim, which is expressly pled in the alternative to the (strong) breach of contract claim.

- The rubric is wrong because a careful litigator could reasonably conclude that a motion to dismiss that claim is not a good idea: Count IV is pleaded in the alternative (Complaint ¶127) and seeks restitution of “no less than \$22,521,312.50” (¶135), which is exactly the Count I contract damages (\$17,721,312.50, ¶94) plus the Count III trade-secret damages (\$4,800,000, ¶120). Counts I and III would survive the motion and double recovery is barred, so dismissing Count IV would not reduce the defendant's net exposure; and the factual underpinnings of both claims are the same so a successful motion would not reduce the scope of discovery. Filing a dispositive motion is costly and, on these facts, there seems to be no benefit to doing so. So the required recommendation would fail many thoughtful analyses that could discuss the pros and cons and recommend against filing such a motion.

5. Draft Conflict Check Memorandum for Litigation Engagement Clearance (C-030)

Criterion. **ISSUE_005: Recommends screening or removal of Caleb Strand.** PASS if the memo recommends that Caleb Strand be screened from the engagement or removed from the proposed staffing. FAIL if no such recommendation is made.

Verdict. Defective (low-quality professional judgment, rubric is incorrect). *Environment.* Correct. *AI auditors' reasoning (reviewer's assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/P; Claude Opus 5.5: P/P.

Analysis. The rubric requires the model to recommend that an associate who shares an apartment with his sister (an in-house IP paralegal at TriPoint, the adverse party) be screened from the matter. Screening is generally used in certain specific situations where required by the professional conduct rules, but no rule requires screening in this scenario.

- I agree that a prudent memo would flag the fact that a junior team member is sharing an apartment with an employee of the adverse party, but that is covered by other rubrics (C-026 through C-029). This rubric specifically fails the work product (and results in a 0 for all work

on this task) even if the model recommended, say, obtaining the client’s informed consent, which would be perfectly permissible and appropriate here.

- Other rubrics in this task resolve similar kinds of conflicts with informed consent (C-024 accepts “clearance with conditions (such as obtaining informed consent)” for Lisa Chow’s spousal conflict, and C-016 recommends informed consent for the Ridgeline conflict). Accordingly, informed consent plus a home confidentiality protocol could suffice as an alternative to screening.

6. Draft Conflict Check Memorandum for Litigation Engagement Clearance (C-049)

Criterion. **ISSUE_011: Recommends integrating annual disclosure questionnaires with ConflictTracker.** PASS if the memo recommends that annual conflict-disclosure questionnaires be cross-referenced with or integrated into the ConflictTracker system. FAIL if no recommendation about integrating disclosure questionnaires with the conflicts database is made.

Verdict. Defective (requires specific outputs not expressly or impliedly requested in the prompt).

Environment. Correct. *AI auditors’ reasoning (reviewer’s assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: F/P; Claude Opus 5.5: P/P.

Analysis. The assignment in the prompt is to draft a “comprehensive conflict check memorandum for the proposed engagement.” The rubric requires that the memo contain suggested firm-wide process improvements (memo must “recommend[] that annual conflict-disclosure questionnaires be cross-referenced with or integrated into the ConflictTracker system”). That’s simply not the assignment, and there’s many reasons why it’s undesirable to jam broad process suggestions into a conflicts memo for one specific case. This rubric actually failed work product from Luna that did everything a clearance memo needs on this point. The two judges split on that same Luna memo (Sonnet 4.6 failed it; GPT-5.5 passed it), which shows the criterion does not grade reliably.

7. Draft Counterclaim Against Plaintiff for Breach of Joint Development Agreement (C-045)

Criterion. **Count: Patent Infringement (ISSUE_005).** PASS if the counterclaim includes a separate count for patent infringement under 35 U.S.C. § 271, alleging that the LumiSense 400 infringes U.S. Patent No. 11,234,567 covering the SensorCore™ architecture. FAIL if no patent infringement counterclaim is asserted.

Verdict. Arguable. *Environment.* Defective: the record lacks the patent the counterclaim must plead, and the two expert reports give different titles for it. *AI auditors’ reasoning (reviewer’s assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/P; Claude Opus 5.5: P/P.

Analysis. The task is to draft a counterclaim based on a handful of documents (including an expert report, some emails, and a counterclaim strategy memo that another lawyer prepared). The rubric dings the AI if the courtclaims do not include “a separate count for patent infringement under 35 U.S.C. § 271, alleging that the LumiSense 400 infringes U.S. Patent No. 11,234,567.”

- The environment does not contain the allegedly infringed patent, which one would need to review in order to have a sufficient basis for drafting that counterclaim, so this environment appears to be incomplete. That said, a defensible approach might be for the agent to draft the counterclaim but flag in brackets that it requires verification.

8. Draft Counterclaim Against Plaintiff for Breach of Joint Development Agreement (C-046)

Criterion. **Patent Infringement Count: Patent Number Cited.** PASS if the patent infringement count specifically identifies U.S. Patent No. 11,234,567. FAIL if the patent number is omitted or incorrect in the patent infringement count.

Verdict. Arguable. *Environment.* Defective: same task environment as item 7. *AI auditors' reasoning (reviewer's assessment).* Partly correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/P; Claude Opus 5.5: P/P.

Analysis. Same issue as prior rubric. This one is essentially derivative because it depends on pleading the patent count that was required in the prior rubric.

9. Draft Case Assessment Memorandum for Defective Industrial Equipment Product Liability Claim (C-003)

Criterion. **Executive Summary mentions death of Marco Reyes.** PASS if the Executive Summary mentions the death of Marco Reyes. FAIL if the fatality is not mentioned in the Executive Summary.

Verdict. Defective (requires specific outputs not expressly or impliedly requested in the prompt). *Environment.* Defective: same task environment as item 10 (Ohio state agency issuing OSHA citations). *AI auditors' reasoning (reviewer's assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/F; Claude Opus 5.5: F/F.

Analysis. The case involved a (fictitious) safety incident at a manufacturing plant that killed one employee and seriously injured two others. The rubric requires the fact that one employee died be discussed specifically in the “executive summary.” When we tested Opus 5.5 and GPT-6-Luna on this benchmark, both discussed the death in several places in their memo, especially the factual background section, but the executive summary consisted of an overview of the legal analysis that did not restate the fact of the death. They were failed by one or more judges.

- This fails perfectly competent and appropriate work product. The company CEO who retained counsel to advise on this incident is the same person who delivered news of the (fictitious) tragic death to the decedent’s widow and surviving family members, so the most senior management of the company is obviously very familiar with the incident and the agents’ approach of focusing the executive summary on the legal exposure is appropriate.
- The two judges reached opposite results on Luna’s memo: GPT-5.5 failed it, while Sonnet 4.6 passed it only by quoting a sentence from the memo’s facts section as if it appeared in the executive summary.

10. Draft Case Assessment Memorandum for Defective Industrial Equipment Product Liability Claim (C-026)

Criterion. **OSHA LOTO citation identified with specific violation and penalty.** PASS if the memo identifies the OSHA serious citation for LOTO deficiency under 29 CFR 1910.147(c)(4)(i) with a penalty of \$18,900. FAIL if this specific citation, regulation, or penalty amount is not referenced.

Verdict. Not defective (defective environment, but rubric does not fail correct work product).

Environment. Defective: Ohio’s Division of Safety & Hygiene cannot issue OSHA citations to this private employer. The criterion itself does not misgrade work (environment-only defect). *AI auditors’ reasoning (reviewer’s assessment).* Wrong. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: F/F; Claude Opus 5.5: P/P.

Analysis. Although the specific rubric being tested does not fail work product incorrectly, the environment is defective in an unintended way. The environment contains an OSHA citation from Ohio’s Division of Safety & Hygiene. But Ohio does not have an OSHA-approved State Plan, so only federal regulators have jurisdiction to issue OSHA citations against this company. The environment does not test that so it is not being presented as a “mistake” by one’s adversary that the agents should pick up on.

- The rubric itself is not wrong because it simply wants a memo to address the purported citation, so an agent could write the memo while noticing that the task is obviously a fictitious evaluation with an incoherent environment.

11. Draft Discovery Plan Memorandum for Breach of Contract and Fraud Defense (C-036)

Criterion. **Provides custodian priority rankings.** PASS if the memo provides priority rankings or tiers for custodians (e.g., high/medium/low priority, or Tier 1/Tier 2, or similar prioritization). FAIL if custodians are listed without any indication of relative priority.

Verdict. Not defective. *Environment.* Correct. *AI auditors’ reasoning (reviewer’s assessment).* Wrong. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/P; Claude Opus 5.5: P/P.

Analysis. The AI agent is right that the rubric is really enforcing a stylistic choice, but the choice here is well-grounded in convention and an appropriate expectation for quality work product. The case file contains a court order where the judge encouraged the parties to discuss whether phased discovery would be appropriate at the Rule 26(f) conference, and there is some existing prioritization work in the environment. The rubric requires that the agent’s discovery plan include some kind of prioritization or phasing of custodians, which is a common expectation in these cases. I think it’s fair to judge anything that lacks any form of prioritization as deficient given the context.

12. Draft Federal Complaint for Breach of Contract and Fiduciary Duty — Placement Agent Dispute (C-025)

Criterion. **ISSUE_012: Complaint requests injunctive relief.** PASS if the prayer for relief includes a request for preliminary and/or permanent injunctive relief — specifically to enforce the non-solicitation provision and prevent Graydon from continuing to divert Fund III investor prospects to competing funds. FAIL if no injunctive relief is requested.

Verdict. Not defective. *Environment.* Correct. *AI auditors’ reasoning (reviewer’s assessment).* Wrong. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/P; Claude Opus 5.5: P/P.

Analysis. The client intake memo specifically describes an interest in seeking injunctive relief (“We may also wish to seek preliminary and permanent injunctive relief to enforce the twelve-month non-solicitation tail”), and the client’s demand letter threatens it. While it’s true that the intake memo flags the topic for discussion, the correct approach is to include the proposed prayer for relief

in the draft, so the case team can see how it reads and discuss any downsides. As a practical matter, there's unlikely to be much downside to including this prayer for relief in the complaint. (Whether it's worth the cost of actually pursuing a preliminary injunction on this is debatable, so the rubric's approach makes sense.)

13. Draft First Set of Interrogatories to Defendant Veridian Health Systems in Trade Secret Misappropriation Case (C-008)

Criterion. **Definitions section defines 'Document' or 'Documents'.** PASS if the Definitions section includes a definition of 'Document' or 'Documents' that is broad enough to encompass electronically stored information (ESI), emails, and physical documents. FAIL if there is no definition of 'Document' or it is so narrow as to exclude electronic records.

Verdict. Defective (unambiguous objective error, rubric is incorrect). *Environment.* Defective: the supplied documents mischaracterize or miscite case law (citation sweep; peripheral to this criterion). *AI auditors' reasoning (reviewer's assessment).* Partly correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: F/F; Claude Opus 5.5: P/P.

Analysis. The rubric fails any draft that does not supply a definition of "document" encompassing electronically stored information (ESI). There are two problems:

- First, these are interrogatories, not document requests. Well-drafted interrogatories do not necessarily need to use the word "document." The rubric does not really make sense as a requirement for this kind of work product.
- Second, even if we were dealing with document requests here (which we are not), the local rules of many courts, including the Western District of Texas (where this fictitious case is being litigated), define the term "document" for all discovery requests served by litigants in those cases. There is thus no requirement to define the term "documents," and in fact, supplying a definition could just trigger objections from the other side that the local rule controls. See Local Rule CV-26, available at <https://www.txwd.uscourts.gov/court-information/lcr-civil-rules/> ("The term 'document' means any document or electronically stored information as described in Federal Rule of Civil Procedure 34(a). A draft of a nonidentical copy is a separate document within the meaning of this term.").
- The rubric does demonstrably fail rules-compliant work product. For example, Luna was failed for not including a definition of the term "Document" but that's actually what the local rules encourage in this Court.

14. Draft First Set of Interrogatories to Defendant Veridian Health Systems in Trade Secret Misappropriation Case (C-026)

Criterion. **Interrogatory re: USB drive or CMT files on Veridian systems.** PASS if at least one interrogatory asks whether Oshiro's USB drive (referencing serial number CX-8827491 or otherwise describing the external storage device) was ever connected to, accessed from, or present on any Veridian system, device, or network, or asks whether any CMT files or materials were ever possessed, accessed, viewed, or copied by Veridian or its employees. FAIL if no interrogatory addresses the USB drive or CMT files on Veridian systems.

Verdict. Not defective. *Environment.* Defective: same task environment as item 13 (environment-only defect). *AI auditors' reasoning (reviewer's assessment).* Wrong. *Native grades* (Sonnet

4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/P; Claude Opus 5.5: P/P.

Analysis. This one does not really belong on the list, as GPT-6-Sol was complaining about a different rubric (C-048) which it argues is inconsistent with this one. It does not identify any issue in this rubric, standing alone, so this counts as an instance where Sol added something to the “flagged” list that does not belong.

15. Draft Discovery Responses and Objections to RFAs and RFPs in Breach of Supply Agreement Litigation (C-028)

Criterion. ISSUE_011: Identifies RFA No. 20 as calling for a legal conclusion. PASS if the plan identifies that RFA No. 20 (asking Prismavale to admit the products were ‘not merchantable as defined by the Uniform Commercial Code’) calls for a legal conclusion regarding UCC § 2-314 merchantability and recommends objecting on that basis. FAIL if the plan does not identify the legal conclusion problem with RFA No. 20.

Verdict. Defective (low-quality professional judgment, rubric is incorrect). *Environment.* Defective: the supplied Answer cites complaint paragraph ranges that do not match the complaint (e.g. Count III as ¶¶ 88–94; the complaint’s Count III is ¶¶ 86–96). *AI auditors’ reasoning (reviewer’s assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: F/F; Claude Opus 5.5: P/P.

Analysis. Plaintiff served a request for admission on the defendant that says, “Admit that the SB-102 and PS-302 products shipped by Prismavale to Greenleaf were not merchantable as defined by the Uniform Commercial Code.” The prompt asks the agent to “prepare a discovery strategy memo and formal RFA/RFP responses.” The rubric requires “the plan” (i.e., the strategy memo) to recommend objecting to the request on the grounds that it calls for a legal conclusion. The rubric is wrong for two main reasons:

- First, the modern view is requests for admissions can call for application of law to facts. While it is within acceptable professional norms for a lawyer to object on the ground that such a request calls for a legal conclusion, and then proceed to answer, the better reading of the law is that this request is not objectionable such that the party should simply answer. On this legal conclusion in particular, there’s really no reason why one would prefer to object rather than simply deny the requested admission. So this rubric fails competent work product (and probably the better answer than the one the rubric calls for).
- Second, even assuming one should include the objection at all, the task fails the model if the issue is not expressly addressed in the strategy memo. Luna made the “legal conclusion” objection in the draft responses and then proceeded a substantive denial (an acceptable approach), but did not discuss the “legal conclusion” objection in the memo. That was judged a failure. That’s really requiring duplicative responses on a very minor point that probably shouldn’t be in the drafts at all.

16. Draft Motion in Limine to Exclude Expert Testimony and Prejudicial Evidence in Commercial Breach of Contract Case (C-009)

Criterion. ISSUE_004: Identifies the \$11,000 arithmetic discrepancy. PASS if the motion identifies that the sum of Marchetti’s five itemized damages categories totals \$14,211,000, but her report states total damages as \$14,200,000, creating an \$11,000 discrepancy. FAIL if this arithmetic error is not identified.

Verdict. Defective (low-quality professional judgment, rubric is incorrect (a judgment call, not an objective error); defective environment (directs advocacy outside professional norms); companion rubric C-010 also incorrect (not sampled)). *Environment.* Defective: the task directs advocacy outside professional norms (an \$11,000 rounding point offered as a *Daubert* ground) and labels a rounding difference an “arithmetic error.” *AI auditors’ reasoning (reviewer’s assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/P; Claude Opus 5.5: P/P.

Analysis. The damages expert includes a summary table in her report listing various items of damages, which collectively add up to \$14,211,000; the Total Damages line lists the total as \$14,200,000. Opposing counsel questioned the expert about it at her deposition, and the expert calls it “a rounding discrepancy.” An internal strategy memo highlights the issue as a ground for seeking to exclude the expert’s testimony under *Daubert*, arguing “the arithmetic error supports a cumulative argument that the report was prepared without the rigor required of expert testimony in federal court.”

- This task is poorly specified, and the recommended strategy of raising this with the judge is a poor one:
- First, the task environment repeatedly states that this is an “arithmetic error.” That is false. The expert has rounded the total in the table to the nearest \$100,000. While that is somewhat unorthodox in an expert report (particularly since it is written out as \$14,200,000 not \$14.2 million), numbers often are rounded in court filings so the judge is unlikely to agree that this is an error.
- Second, even if there were an error (which there is not), that is generally not a basis for excluding the expert’s testimony under *Daubert*. At most one could move to exclude the erroneous version and ask for more accurate testimony, but that does nothing to help the hypothetical client here; it just raises their damages exposure by the \$11,000 that the expert was willing to round off.
- Third, the proposed framing of the argument comes very close to a breach of decorum and carries a real risk of drawing a rebuke from the judge. The strategy memo wants the motion to argue the expert’s 30 prior engagements were “in state courts and arbitrations that do not impose the same methodological gatekeeping requirements” as federal courts. While some state courts use the *Frye* standard instead of *Daubert*, many federal judges would take offense at the suggestion that state courts do not observe adequate methodological rigor. Many busy judges could be annoyed at spending time on an argument that literally just amounts to a rounding error.
- This rubric is scored as Defective, as a professional-judgment call rather than an objective error. Our view is that the better approach is to drop this issue entirely, and the sampled rubric (C-009) fails a motion that does so, even though a motion that merely mentions the discrepancy in a footnote would pass. The companion rubric, C-010, is in my view objectively wrong because it requires that this issue be presented as a basis for excluding the expert’s testimony under *Daubert*. Luna was failed incorrectly on C-010 (by the Sonnet 4.6 judge) for downplaying this issue; in fact, Luna was right to do so, as it should not be raised at all. But since that’s not the sampled criterion, we do not mark it here.

17. Draft Rule 12(b)(6) Motion to Dismiss Brief — Commercial Software Licensing Dispute (C-037)

Criterion. **ISSUE_008 — Cites Delaware parol evidence authority.** PASS if the motion brief cites relevant Delaware authority on the parol evidence rule and integration clauses, such as *SIGA Technologies v. PharmAthene* or *Eagle Industries v. DeVilbiss Health Care*. FAIL if no Delaware parol evidence authority is cited.

Verdict. Defective (unambiguous objective error (including hallucinations), rubric is incorrect).

Environment. Defective: the supplied research memo cites fabricated or mischaracterized authority (citation sweep: 28 of 53 citations in this task have problems). *AI auditors' reasoning (reviewer's assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: F/F; Claude Opus 5.5: F/F.

Analysis. The rubric here involves a legal error, and the task environment contains hallucinations about case law. This one is extremely defective for multiple reasons:

- First, the rubric is demanding case law on the wrong legal doctrine. Under Delaware law, the parol evidence rule and an integration clause do not preclude fraud claims. The issue here is whether the plaintiff can reasonably allege that they relied on their counterparty's pre-signing representations despite their agreement in the contract that they were not relying on any representations other than those written in the contract itself.
- Second, the cases in the environment appear to contain multiple hallucinations. The rubric refers to *SIGA v. PharmAthene*, but Opus noted that case does not address integration or parol evidence. The task environment contains a legal research memo that cites *SIGA Technologies, Inc. v. PharmAthene, Inc.*, 67 A.3d 330 (Del. 2013), and claims that it states (at page 344), "an integration clause is the clearest declaration that the parties intend the writing to be the complete and final expression of their agreement." The quoted language does not appear in the cited case. Page 344 of that case addresses the duty to negotiate in good faith, not integration clauses. Similarly, the memo cites *Eagle Industries, Inc. v. DeVilbiss Health Care, Inc.*, 702 A.2d 1228 (Del. 1997) and says: "The Delaware Supreme Court held that 'a party to a contract cannot promise, in a clear integration clause of a negotiated agreement, that it is not relying on promises or representations made by the other party outside of the agreement, and then assert a claim for fraud based on those very representations.' *Id.* at 1232." The quoted language does not appear in the cited case.
- Third, the rubric failed responses that stated the correct law on this point. In our tests, Opus reviewed *SIGA* and *Eagle*, correctly realized they were inapposite and declined to cite them, cited more on-point and accurate authority, and was then failed for doing a better job than the rubric's author did at this task.

18. Draft Rule 12(b)(6) Motion to Dismiss Brief — Commercial Software Licensing Dispute (C-045)

Criterion. **ISSUE_010 — References Arcadia's own delays contributing to Go-Live delay.** PASS if the motion brief identifies Arcadia's own contributing failures, including the 45-day Project Manager gap (Kevin Liu's departure) and/or the 75-day late API specifications, as causes of the Go-Live delay. FAIL if Arcadia's own contributory delays are not mentioned.

Verdict. Defective (unambiguous objective error, rubric is incorrect). *Environment.* Defective: same task environment as item 17. *AI auditors' reasoning (reviewer's assessment).* Correct. *Native*

grades (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: F/F; Claude Opus 5.5: F/F.

Analysis. The rubric makes a rookie mistake, demanding that the agent argue a factual dispute on a motion to dismiss, where the court is required to assume the plaintiff's allegations are true and cannot consider factual dispute. The 45-day and 75-day delays complained of come from the defendant's internal report, which the plaintiff's complaint disputes. Moreover, courts generally cannot resolve messy fact disputes over "causation" on a motion to dismiss.

- The criterion failed agents that correctly recognized that this argument was outside the scope of what can be argued on a motion to dismiss and thus decided to excluded it.

19. Draft Verified Responses and Objections to Plaintiff's First Set of Interrogatories in Commercial Breach of Contract and Fraud Action (C-028)

Criterion. Correct financial figures for Tri-Basin purchase history. PASS if any purchase history figures cited in the responses are consistent with the canonical data: 2015: \$9.2M; 2016: \$10.1M; 2017: \$10.8M; 2018: \$12.4M; 2019: \$13.6M; 2020: \$9.7M; 2021: \$15.3M; 2022: \$16.8M; 2023 (Jan–Aug): \$11.9M. Minor rounding differences are acceptable. Also PASS if the response invokes Rule 33(d) rather than citing specific figures. FAIL if specific purchase figures are cited but are materially incorrect.

Verdict. Defective (unambiguous objective error, deficiency in task environment; rubric is incorrect).

Environment. Defective: the purchase-history spreadsheet is internally inconsistent (three 2020 series; line items that do not sum to subtotals), and the record has no contract-year data. *AI auditors' reasoning (reviewer's assessment).* Partly correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/P; Claude Opus 5.5: P/P.

Analysis. There are two issues:

- First, Interrogatory No. 4(b) asks for Tri-Basin's actual purchases for the 2020 contract year (the contract year runs from January 15 through January 14 of the following year; Interrogatory No. 14 separately asks for calendar-year totals), but the canonical data referred to in the rubric is calendar year data, so the rubric's numbers are not correct for that request. The correct value cannot be discerned from the environment: the monthly data cannot be split at January 15, and nothing in the record supports assuming purchases are spread evenly within January (seasonal variation would make that unlikely). The rubric therefore grades wrong in the opposite direction from most of the other items: rather than failing good work product, it passes responses that supply the calendar-year figure for a contract-year request without acknowledging that the record does not contain the requested data. That is approving fundamentally deficient work product.
- Second, the data that is provided has some internal inconsistencies. For example, an "annual summary" tab lists 2020 purchases at \$9.7 million, but the monthly detail lists 2020 three times, with totals of \$8,700,000, \$8,549,400 and \$9,700,000. The units sold do not match up between those two sheets for 2021, 2022, 2023, and series 3 of 2020. One of the monthly rows does not add up (gross \$1,210,000 + credits $-\$17,400 = \$1,192,600$, but the net shown is \$1,192,500). The annual summary minimums do not add up. In the Product Line Breakdown, line-item net revenue never sums to the subtotal. So a response that relied on the line items would get graded as a failure.
- The environment contains law firm discovery guidelines that recommend that interrogatory responses match the client's Answer, but the Answer here contains only approximate figures,

which is consistent with the Answer using calendar year data as a rough approximation instead of providing the contract year data requested.

20. Extract Key Obligations from Litigation Hold and Document Preservation Notice — Obligation Summary Memorandum (C-002)

Criterion. **ISSUE_001: Flags financial records date discrepancy as requiring DOJ clarification.** PASS if the memo recommends seeking clarification from the DOJ regarding the conflicting date ranges (Jan 1, 2017 vs. Jan 1, 2019) for financial records. FAIL if no recommendation to seek clarification is made.

Verdict. Defective (unambiguous objective error, rubric is incorrect). *Environment.* Correct. *AI auditors' reasoning (reviewer's assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: F/F; Claude Opus 5.5: P/F.

Analysis. The task involves a document preservation notice from federal criminal authorities that defines the relevant period (for which the recipient must preserve documents) as January 1, 2019, through March 3, 2025 “unless otherwise specified herein.” For one specific category of documents, the notice states that, “Notwithstanding the Relevant Period defined in Paragraph 12 of this Notice, Ridgeline shall preserve [certain kinds of] records . . . from January 1, 2017 to the present date. ”

- The rubric suggests that the agent should seek clarification of the “conflicting” date ranges but there is no conflict. The notice quite clearly imposed a longer period for one specific category of documents, while all the other categories are subject to a shorter period. This is a fairly common request.
- All the models were failed on this criterion, but the criterion makes no sense.

21. Identify Excessive or Duplicative Research Charges in Litigation Invoice (C-013)

Criterion. **Issue 3: Identifies duplicative MTCA research across Takahashi and Wendt.** PASS if the memo identifies that multiple attorneys (Kenji Takahashi and Tyler Wendt) billed for overlapping MTCA liability/contractor liability/summary judgment research during July 2024, citing Section 6.4 (Duplicative Research Prohibition). FAIL if the MTCA duplication across timekeepers is not identified.

Verdict. Not defective. *Environment.* Correct. *AI auditors' reasoning (reviewer's assessment).* Wrong. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/P; Claude Opus 5.5: P/P.

Analysis. While Sol is right that the facts here are insufficient to establish that the client’s billing guidelines were violated, for purposes of the task (a client-side memo identifying changes to dispute). it seems reasonable to flag that multiple attorneys were doing research into closely related issues. The task requested a memo “a memo identifying all non-compliant or excessive charges with recommended adjustments,” so flagging potential violations (even if they were not clearly proven from the face of the time entries) is a standard approach a client would take here.

22. Identify Issues in Litigation Matter Budget Proposal (C-029)

Criterion. **Budget inconsistency rated as Critical or High severity.** PASS if the fee total reconciliation discrepancy (\$116,050 difference between phase-based and staffing-model totals) is

assigned a severity of Critical or High (or the highest or second-highest tier in whatever severity scale is used). FAIL if it is assigned a lower severity level or no severity level.

Verdict. Arguable (requires specific outputs not expressly or impliedly requested in the prompt). *Environment.* Correct. *AI auditors' reasoning (reviewer's assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/P; Claude Opus 5.5: P/P.

Analysis. The prompt requested “a categorized issue memorandum” but didn’t specify what the categories should be. The rubric fails the model if it does not break issues into severity levels or it does not assign a critical or high severity level to a specific issue. While it appears that models often do volunteer this kind of severity assessment, it’s not requested by the prompt and competent work product could decline to include it.

23. Review Counterparty’s Proposed Jury Instructions — Issue Memorandum for Trade Secrets Trial (C-020)

Criterion. **Explains distinction between tortious interference with contract vs. business relations.** PASS if the memo explains that under Georgia law, tortious interference with an existing contract does not require a separately/independently wrongful act — the knowing inducement of the breach is sufficient — and that the ‘independently wrongful act’ element applies to the distinct tort of tortious interference with business relations. FAIL if this legal distinction is not articulated.

Verdict. Defective (unambiguous objective error (including hallucinations), rubric is incorrect).

Environment. Defective: the fictional summary-judgment order cites a case for the opposite of its holding and a case that cannot be found (citation sweep: 10 of 21 citations in this task have problems).

AI auditors' reasoning (reviewer's assessment). Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: F/F; Claude Opus 5.5: F/F.

Analysis. Multiple issues:

- First, this is a clear legal error. Georgia applies the same elements to interference with contractual and business relations. The rubric says the opposite of what Georgia law is. *Disaster Services, Inc. v. ERC P’ship*, 228 Ga. App. 739, 740–41 (1997) (“Tortious interference claims, whether asserting interference with contractual relations, business relations, or potential business relations, share certain common essential elements: (1) improper action or wrongful conduct by the defendant without privilege; (2) the defendant acted purposely and with malice with the intent to injure; (3) the defendant induced a breach of contractual obligations or caused a party or third parties to discontinue or fail to enter into an anticipated business relationship with the plaintiff; and (4) the defendant’s tortious conduct proximately caused damage to the plaintiff.”).
- Second, the fictional summary judgment order (from which the rubric’s legal position seems to be taken) appears to contain several hallucinations. For example, the order paraphrases the third and fourth element described in *Disaster Services* approximately correctly, but the first two elements are erroneously listed as “(1) the existence of a valid contract between the plaintiff and a third party; (2) the defendant’s knowledge of the contract.” The fictional order also cites *Valdosta Livestock, Inc. v. Furst*, 342 Ga. App. 25, 28 (2017), which is not found in any legal databases.

24. Review Litigation Invoice Against Outside Counsel Billing Guidelines — Compliance Deviation Report (C-024)

Criterion. **Recommends full disallowance of black car service: \$387.00.** PASS if the report recommends disallowing the full \$387.00 for black car service (or reducing it to a reasonable taxi/ride-share equivalent). FAIL if no disallowance or reduction is recommended.

Verdict. Not defective. *Environment.* Correct. *AI auditors’ reasoning (reviewer’s assessment).* Wrong. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/P; Claude Opus 5.5: P/P.

Analysis. The billing guidelines state that “Reimbursable ground transportation is limited to ride-share ... or taxi” and “Limousine services and black car services are not reimbursable under any circumstances.” Sol’s critique of the rubric is that the policy requires disallowing black car reimbursement, not reducing the reimbursement amount to what a taxi or ride share would have cost. The rubric’s interpretation (that either disallowing the expense entirely or capping reimbursement at the amount a taxi would have cost) seems reasonable and closer to the likely real-world application of a policy like this.

25. Review Litigation Invoice Against Outside Counsel Billing Guidelines — Compliance Deviation Report (C-042)

Criterion. **Classifies business class airfare as borderline/requires discussion.** PASS if the report classifies the business class airfare issue (Messina’s \$2,847.00 flight) as a borderline or ‘requires discussion’ issue rather than a clear-cut mandatory reduction, given the 4-hour-20-minute flight duration and the 4-hour threshold. FAIL if it is classified as a clear mandatory reduction without acknowledging the ambiguity, OR if it is entirely dismissed as compliant.

Verdict. Defective (unambiguous objective error, rubric is incorrect). *Environment.* Correct. *AI auditors’ reasoning (reviewer’s assessment).* Correct. *Native grades* (Sonnet 4.6/GPT-5.5; P pass, F fail): GPT-6 Luna: P/P; Claude Opus 5.5: P/P.

Analysis. The rubric’s analysis is wrong. The travel policy included in the environment states, “For domestic flights with a scheduled duration of four (4) hours or more, business class may be booked without prior approval.” The record does not state the flight’s duration (the invoice lists only the route and fare class). The only basis for raising this seems to be in the rubric itself, which says the flight could take 4h20m, but even under that view, there’s nothing to discuss, as the policy allows business class for that duration.

D A prospective test of outcome feedback

A follow-up would freeze a base model and an earlier training cohort, then compare outcome feedback with a matched control on later, disjoint cases. Related cases should remain together across splits. The primary endpoint would be a predeclared difference in proper score; calibration, failure rates, cost, and reasoning validity would be secondary endpoints. Blinded expert assessment of advice or drafting would test transfer.

For a valid forecast, the implemented Brier-shaped reward is

$$R(p, Y) = 0.02 + 0.98[1 - (p - Y)^2], \quad (6)$$

while invalid output receives zero. The positive affine transformation preserves the proper-score optimum among valid forecasts. The implementation also supplies threshold-based and clipped-log alternatives. Evidence spans are logged rather than rewarded. This design separates a directly scored prediction from an assertion that the accompanying reasoning is legally correct.

For a model emitting a rationale z and forecast p , a candidate objective is

$$\max_{\theta} \mathbb{E}_{(X,U,Y) \sim \mathcal{D}_{\text{train}}, (z,p) \sim \pi_{\theta}(\cdot|X,U)} \left[1 - (p - Y)^2 \right]. \quad (7)$$

This is an outcome-scored prediction objective, not a simulator that allows the agent to alter the litigation trajectory. Optimizing it need not require reinforcement learning: supervised probabilistic prediction, calibration, and policy-gradient approaches are competing methods to test. It also does not directly reward faithful rationales. Leakage-controlled held-out cases and an independent transfer endpoint are therefore necessary to distinguish generalization from memorized case outcomes or improved probability reporting alone.